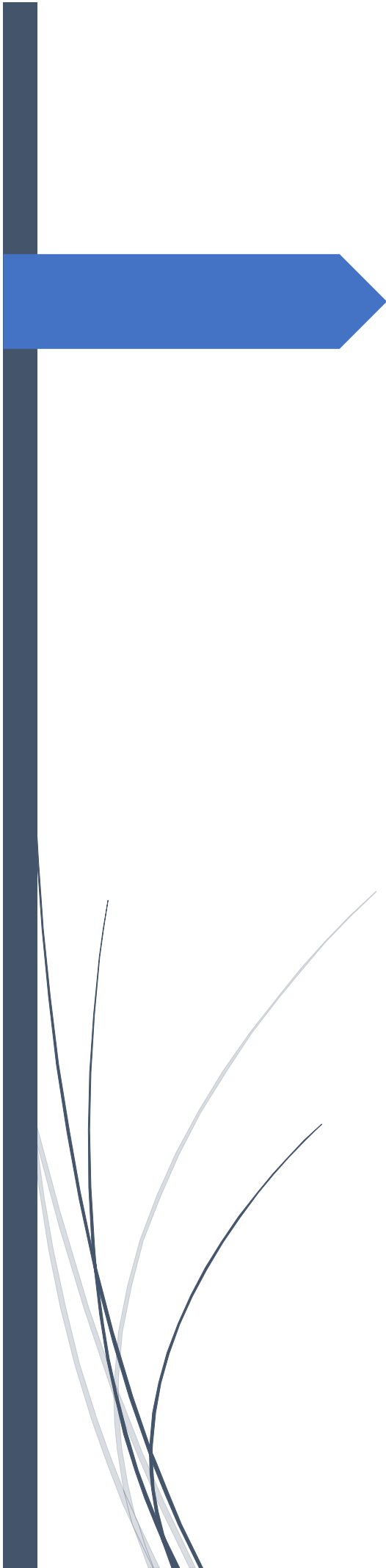




Case Study -5

AI-Powered Document Assistant

[shravya asam](#)

LEAD BUSINESS ANALYST

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1. Project Summary:

An organization receives a large number of business documents such as invoices, applications, forms, and supporting documents that require manual review and data extraction. The proposed AI-powered document assistant uses AI to extract relevant information, summarize documents, identify missing details, and route them for human review, helping reduce manual effort and improve processing efficiency.

2. Business Problem:

- Large volumes of documents require manual review and data extraction.
- Employees spend significant time reading and entering information into systems.
- Missing or inconsistent information can cause processing delays.
- Different document formats make standardization difficult.
- Manual processing increases operational effort and the risk of human error.
- Business teams need faster access to relevant information for decision-making.

3. AI Use Case:

The platform uses **Generative AI and document intelligence** to support employees in reviewing business documents.

Key AI Capabilities

- Extract important information from uploaded documents.
- Summarize lengthy documents.
- Identify missing or inconsistent information.
- Answer questions based on document content.
- Highlight items that require human review.
- Generate a structured summary for downstream processing.

Example:

Upload Document → AI Reads & Extracts Data → Summarizes / Validates → Flags Exceptions → Human Review → Final Processing

4. Current vs. AI-Enabled Process:

Current Process

Document Received → Manual Review → Data Entry → Validation → Decision / Processing

AI-Enabled Process

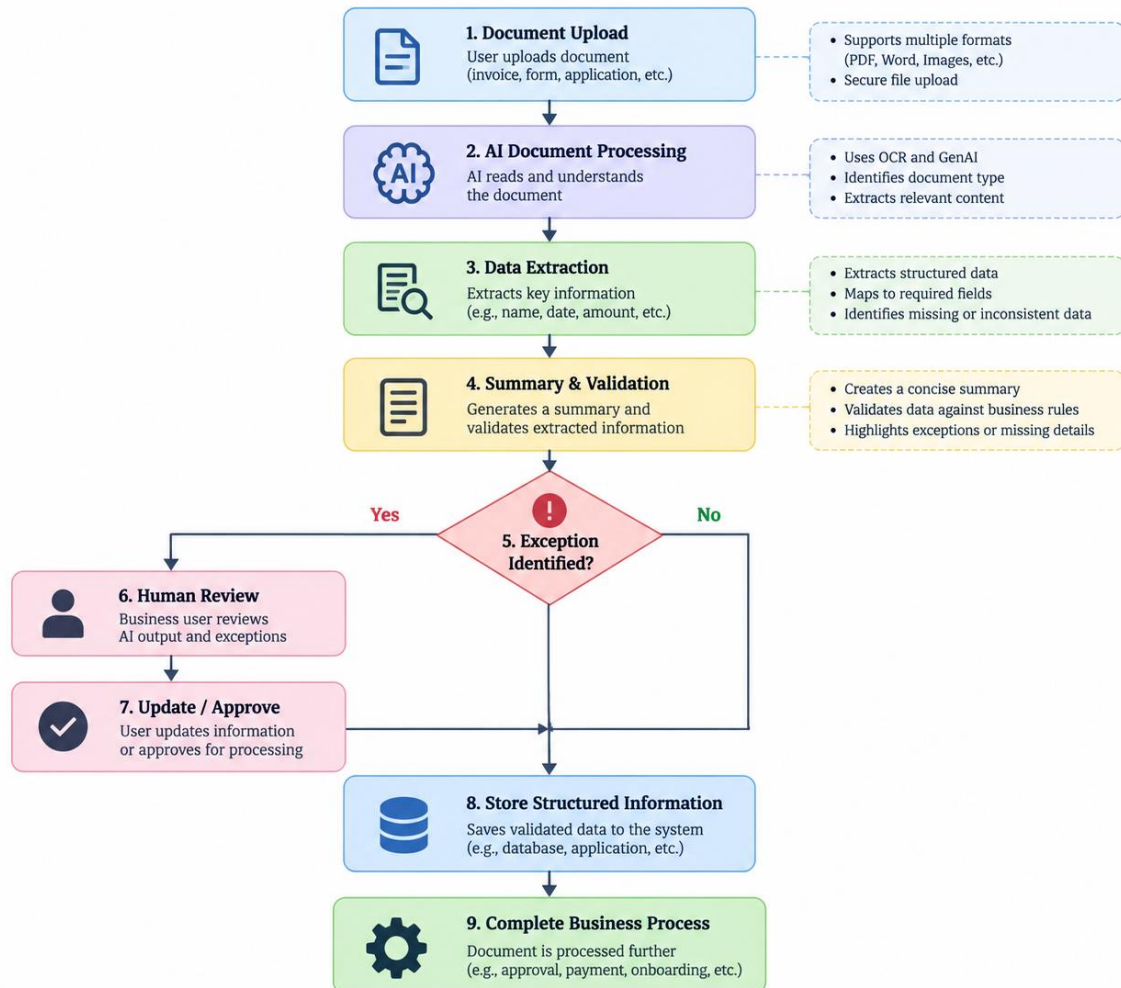
Document Upload → AI Extraction → AI Summary & Validation → Exception Flagging → Human Review → Final Processing

Key Improvement: AI reduces repetitive document review and data-entry effort while keeping human validation for important decisions.

5. AI Workflow:

AI-Powered Document Assistant – Workflow

From document upload to final processing with human-in-the-loop



6. Sample Prompt & AI Output:

Sample Prompt

Review the uploaded invoice and extract the vendor name, invoice number, invoice date, total amount and payment due date. Identify any missing or inconsistent information and provide a short summary.

Illustrative AI Output

Vendor: ABC Supplies
Invoice Number: INV-10458
Invoice Date: 12 Aug 2026
Total Amount: \$5,250
Payment Due Date: 11 Sep 2026

Summary: Invoice contains the required payment information. No missing mandatory fields were identified.

Validation: Human review required before final processing.

Note: Prompt and output are illustrative portfolio examples.

7. Human Review & Validation:

AI-generated information is reviewed by a business user before it is used for final processing.

AI Output → Human Review → Validate Information → Correct if Required → Approve → Final Processing

Validation Checks

- Extracted information is complete and accurate.
- AI summary reflects the source document.
- Missing or inconsistent data is identified.
- Business user can edit incorrect information.
- Final approval remains with the human reviewer.

8. AI Functional Requirements:

FR-001 – Document Data Extraction

The system shall extract predefined information from uploaded documents.

FR-002 – Document Summarization

The system shall generate a concise summary based on the document content.

FR-003 – Data Validation

The system shall identify missing, inconsistent, or potentially incorrect information in the extracted data.

FR-004 – AI Confidence / Validation Status

The system shall display an AI confidence score or validation status for extracted information, where applicable.

FR-005 – Human Review

The system shall allow authorized users to review and validate AI-generated information.

FR-006 – Data Correction

The system shall allow users to edit and correct information identified as inaccurate or incomplete.

FR-007 – Final Validation

The system shall store the final validated information for downstream processing.

FR-008 – Human Approval

The system shall require human validation/approval before AI-generated information is used for final business processing.

9. AI Risk & Control:

Risk	Control
Incorrect AI-generated information	Human review before final processing
Missing information	AI flags incomplete fields
Inconsistent output	Validation rules and structured output
Sensitive document exposure	Role-based access and secure document handling
Incorrect AI summary	User validation against the source document
Over-reliance on AI	Human remains accountable for final decisions

Key Principle:

AI assists the business process; it does not replace human validation for important decisions.

10. AI Evaluation & Success Measures

Metric	Purpose
Extraction Accuracy	Measure accuracy of information extracted from documents
Validation Accuracy	Measure correctly identified missing or inconsistent information
Human Review Rate	Track the percentage of documents requiring manual review
Processing Time	Compare AI-assisted processing with the manual process
Exception Rate	Monitor documents requiring correction or investigation

Success Indicator:

Improve document processing speed and accuracy while maintaining appropriate human oversight.

